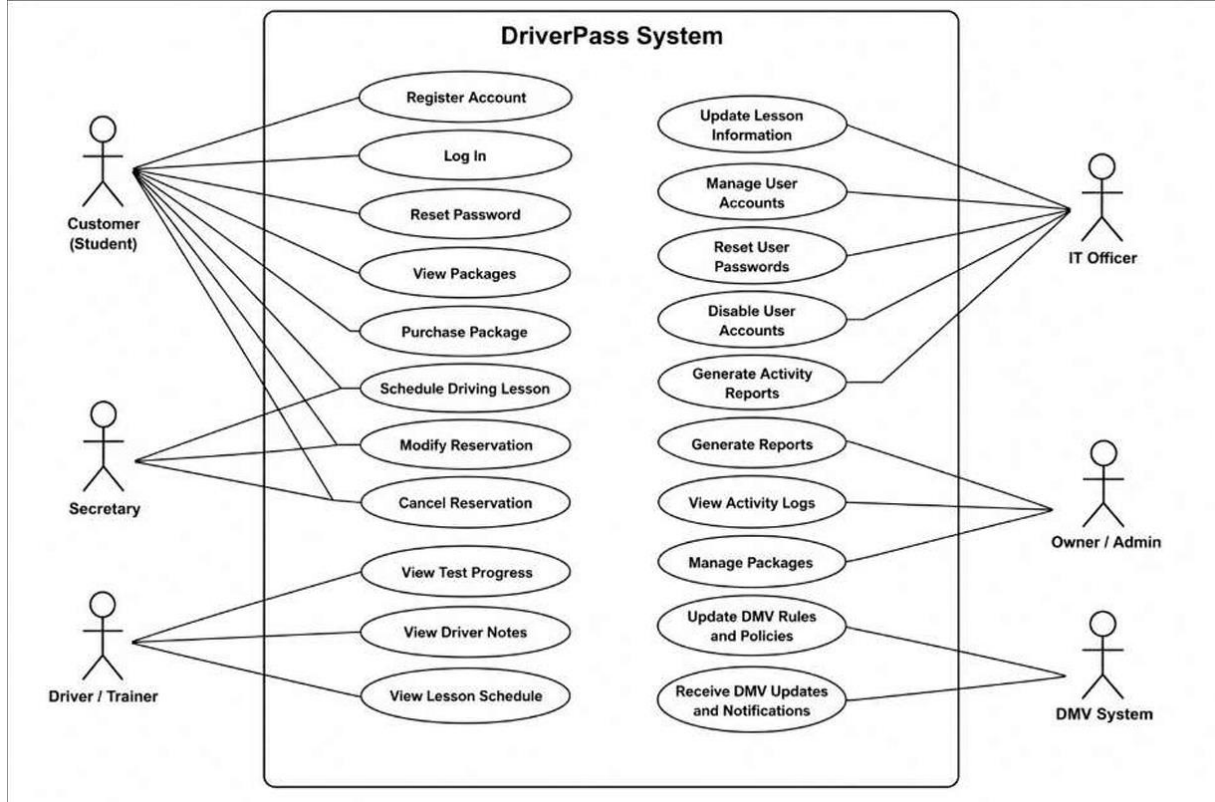


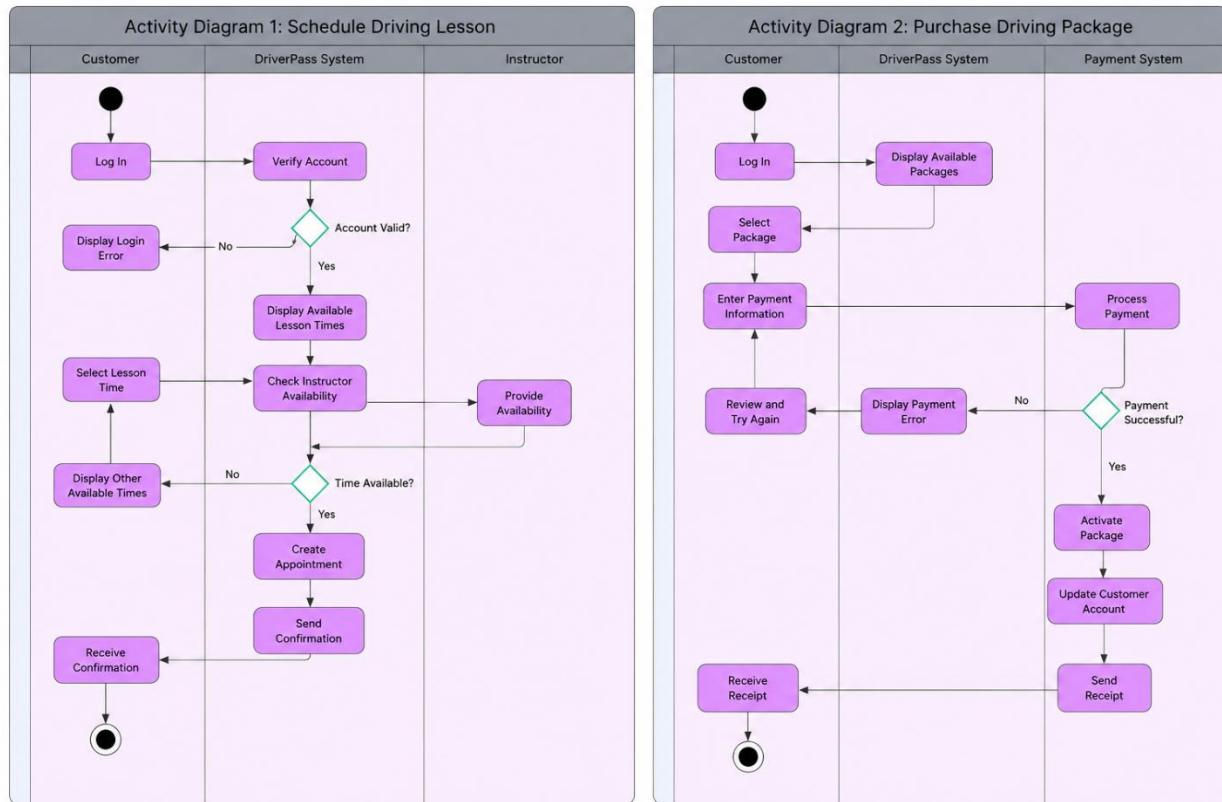
CS 255 System Design Document Template

UML Diagrams

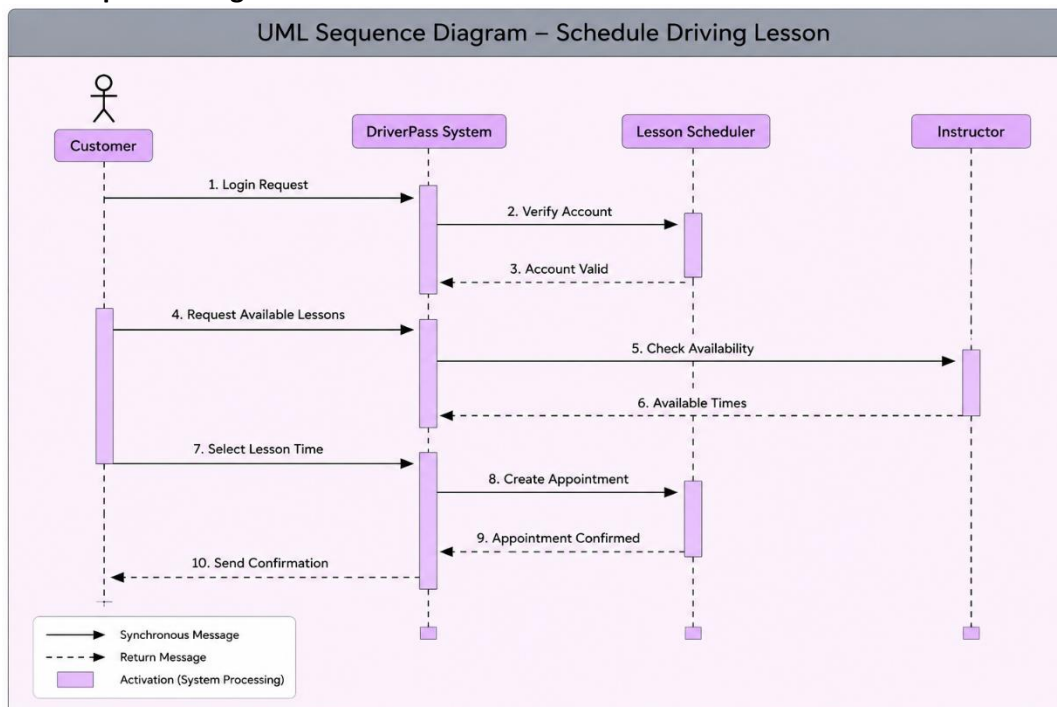
UML Use Case Diagram



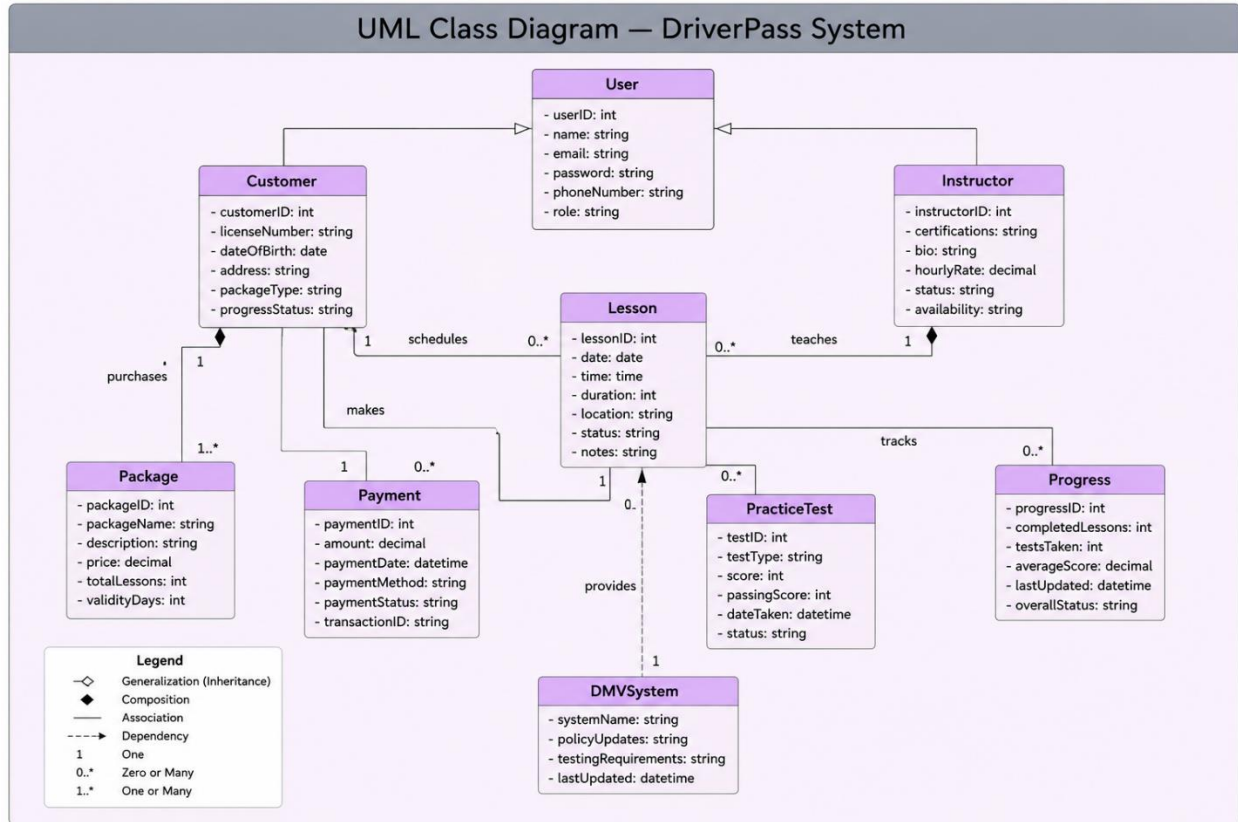
UML Activity Diagrams



UML Sequence Diagram



UML Class Diagram



Technical Requirements

The DriverPass system will need a reliable web-based application that allows customers, instructors, and employees to access the system from different locations and devices. The system will need to have important functions such as customer account creation, lesson scheduling, practice testing, package purchases, and progress tracking. To support these features, the system needs a combination of appropriate hardware, software, security measures, and infrastructure that will allow the system to operate efficiently and provide a positive experience for users.

The hardware requirements for the DriverPass system include devices that allow customers, instructors, and employees to access the application. Customers should be able to use computers, tablets, or mobile devices with an internet connection to access their accounts, schedule lessons, complete practice tests, and review their progress. The system will also require server resources to store application data and manage user requests. These servers may be hosted internally or through a cloud-based environment. The system should also include reliable storage solutions and backup systems to prevent data loss and ensure that important customer information can be recovered if needed.

The software requirements for the DriverPass system include a web application platform, database management system, and tools needed to maintain and operate the application. The system also needs software that can manage user accounts, store customer information, maintain lesson

schedules, process payments, and track student progress. A database will be needed to safely store information such as customer profiles, instructor availability, lesson appointments, practice test results, and package details. The system should also include authentication software to allow users to securely log in and ensure that each user can only access the information appropriate for their role. The system will require secure infrastructure to support communication between users and the application. A hosted server environment will allow the system to remain available while providing the ability to update and maintain the application. Network requirements include a stable internet connection, secure communication between the user and the system, and monitoring tools to identify potential issues. Regular system backups should also be performed to protect information and help maintain availability if technical problems occur.

Security is an important requirement because the DriverPass system will store personal customer information and payment-related data. The system should include secure login procedures, password protection, and encryption to protect user information. Access controls should be implemented so that customers, instructors, and employees only have access to the features and information necessary for their responsibilities. Payment processing should also use secure methods to protect financial information and reduce the risk of unauthorized access.

The DriverPass system will also need ongoing maintenance and support tools to make sure that everything is in proper working order. Developers and administrators will need tools to monitor system performance, identify errors, and apply updates when necessary. The system should be designed so that new features, updated DMV requirements, and additional customer needs can be added in the future without requiring a complete redesign. These technical requirements will help verify that the DriverPass system is reliable, secure, and capable of supporting everyone who uses it.